Linkage of Course Outcome with Assessment Methods and their Weights:

COs	Assessment Method	(100%)
	Class Assessment	
	Attendance	5%
	Class Performance	5%
CO 1-2	Assignment	5%
CO 1-2	Class Test	15%
	Exam	
CO 1-2	Final Exam	70%

Course Title: Engineering Economics
Course Code: GED 221, Credit: 3.0, Total hours: 42

Course Description:

- Introduction to economics.
- Supply and demand
- Various forms of market
- Analysis of costs.
- Analysis of national income.
- Banking- commercial bank and central Bank.
- Government Budgets and Fiscal Policy
- Economic development and growth.

Course Learning Outcomes (CLO):

Upon successful completion of this course, students will be able to:

1. develop preliminary knowledge of economic concepts for understanding of national and international economy.
2. understand how markets work, when markets fail and how the government can affect market outcomes.
3. evaluate macroeconomic policies based on their effects on the economic variables of interest.
4. apply different techniques of investment decisions.

Mapping of Course Outcome to Program Outcome

	PLO1	PLO2	PLO3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11	PLO 12
CLO1										m		
CLO2												
CLO3												
CLO4											s	

Class	**Topics/Assignment**	COs	Reading Reference
1	The Fundamentals of Economics	1,3	1,2
2	Central Problems of an Economy	1	1,2
3	Different Economic systems	1	1,2
4	Production Possibility Curve (or Transformation Curve)	1	1,2
5	Utility, Diminishing marginal utility	1	1,2
6	Theory of Demand	1	1,2
7	Elasticity of demand- price, income and cross elasticity	1	1,2
8	Theory of Supply.	1	1,2
9	Equilibrium	1	1,2
10	Class test -1		
11	Consumer surplus and Producer surplus	1	1,2
12	Indifference curve technique	1	1,2
13	Theory of Production	1	1,2
14	Analysis of cost.	1	1,2
15	Various forms of market- monopoly, perfect competition, monopolistic competition, oligopoly, duopoly.	2	1,2
16	National Income.	1,3	1,2
17	Consumptions, Savings and Investment.	1,3	1,2
18	Inflation and effect of inflation on production and distribution of wealth.	1,3	1,2
19	Types of inflation	1,3	1,2
20	Unemployment	1,3	1,2
21	Class test-2		
22	Banking- Commercial Banking and Central Bank	1	1,2
23	Government Budgets and Fiscal Policy	1,3	1,2
24	Poverty and Economic Inequality	1,3	1,2
25	Economic Growth	1,3	1,2
26	Theories of developments	3	1,2
27	Development partners (World Bank, Asian Development Bank, World Trade Organization, International Monetary Fund)	3	1,2
28	Cost benefit analysis, NPV	4	1,2
29	IRR, Payback period	4	1,2

Reference Book(s):

1. Dewett, K K. and Chand Adarsh, Modern Economic Theory, Shyamlal Charitable Trust, New Delhi, 2003.
2. Samuelson, Paul & Nordhaus, William D., Economics, McGraw-Hill Inc., New York, 1995
3. Parkin, Michael, "Microeconomics", Addison – Wesley Publishing Company.

Assessment Methods:

Category	Marks %
Class Participation	5
Class Attendance	5
Quizzes/class tests	15
Assignment	5
Final Examination (3 hours)	70
Total	**100**

LEVEL- 3 TERM-I

Course Title: Operating System
Course Code: CSE 311, Credit: 3.0, Total hours: 42

Course Description:

1. Introduction of Operating System, Types of OS; Process: process managements, process states, job and process scheduling, CPU scheduling algorithms, process coordination, critical section problems, semaphores, Inter-Process Communication (IPC), classical IPC problems, multiprocessing and timesharing, message and mailbox etc

2. Memory management: swapping, memory allocation schemes, Paging and segmentation, virtual memory, page replacement strategies, working sets, demand paging; Input/output: hardware/software, disk, disk scheduling algorithms, Secondary storage management, terminals, clocks

3. Deadlock: resource allocation, detection, prevention, avoidance and recovery.

4. Main Features of Windows, UNIX, LINUX; Introduction to distributed operating systems.

Course Outcome (CO) Matrix:

At the end of the course, the students should be able to:

		Level of Domain				PO
		C	P	A	S	
CO 1	**Illustrate** the fundamentals concepts of computer system organization and the structure of operating systems.	C2				PO1